Converting Detected Issues into Formal Clinical Queries

Query Generation Module — Clinical Data Management (CDM) Training

Prepared by: Suraj Anil Patil

Guide: Mrs. Nidhi Jha

Forms covered: Baseline Visit, Demographics, Vital Signs

Introduction

In any clinical trial, data collected at site level is rarely perfect on first entry — values may be missing, out of the expected clinical range, entered in the wrong format, duplicated, or logically inconsistent with other data points for the same subject. Before such data can be used for statistical analysis or submitted to a regulatory authority, every discrepancy must be identified, documented, and resolved through a formal, auditable process. In Clinical Data Management (CDM), this process is called query management, and the individual communication sent to a site to resolve a discrepancy is called a query.

This module picks up directly after the data profiling and error-classification stage carried out earlier on the dummy dataset in Castor EDC. Once an issue is detected and classified, it must be converted into a clear, standardised, and professionally worded query before it can be sent to the investigator or site coordinator. Doing this manually for every issue is slow and inconsistent between reviewers, so this task explores a rulebook-driven and AI-assisted approach: a fixed set of rules decides when a query should be raised, a template library keeps the wording standardised, and AI-generated first drafts are reviewed and edited by the CDM team before being released — combining speed with the human judgement needed for GCP-compliant communication.

Main Purpose

The purpose of this task is to convert data issues already detected during data profiling and error classification (missing values, duplicates, invalid formats, outliers, and inconsistent entries) into formal clinical data queries. A query is the standard mechanism by which a Clinical Data Manager communicates a data discrepancy to the site or investigator and requests clarification or correction. This module builds a rulebook that maps each issue type to a query trigger condition, a library of reusable query wording templates, a sample log of AI-drafted queries reviewed and edited for accuracy and tone, and a short explanation of where this step sits in the overall CDM workflow.

Types of Clinical Queries

Queries in CDM generally fall into three origin types, all of which feed into the same rulebook and template system used in this module:

Type	How It Is Raised	Example in This Dataset
System / Edit-Check Query	Auto-generated by the EDC (Castor) when a built-in edit check fires, e.g. a range check or a mandatory-field check.	Systolic BP entered as 310 mmHg triggers an automatic range edit-check.
Manual Query	Raised by a CDM reviewer during manual data review or listing review, based on judgement rather than a coded rule.	Reviewer notices ethnicity free-text spelled two different ways for the same subject.
AI-Assisted Query	A detected issue is matched to a rulebook entry, a draft query is generated automatically from a template, and a human reviewer edits and approves it before release.	All 18 rules in Section 2 of this document use this approach.

Query Rulebook

Each rule below defines the condition that triggers a query, the form/field it applies to, the query category, and the severity used for prioritisation during review.

Rule ID	Form / Field	Trigger Condition	Query Category	Severity
QR-01	Demographics – Age	Age field is blank	Missing Value	Major
QR-02	Demographics – Date of Birth	DOB entered in an invalid or non-standard date format (e.g. text, wrong order)	Invalid Format	Major
QR-03	Baseline – Subject ID	Subject ID is duplicated across two or more records	Duplicate Record	Critical
QR-04	Vital Signs – Systolic BP	Value falls outside the protocol-defined plausible range (e.g. <70 or >250 mmHg)	Out-of-Range / Outlier	Major
QR-05	Vital Signs – Heart Rate	Value is physiologically implausible (e.g. <30 or >220 bpm)	Outlier	Major
QR-06	Baseline – Visit Date	Visit date is earlier than the subject's informed consent / enrolment date	Logical Inconsistency	Critical
QR-07	Demographics – Gender	Gender field contains a value outside the approved coded list (e.g. free text, typo)	Invalid Format	Minor
QR-08	Baseline – Height / Weight	Height or weight recorded as zero or a non-numeric character	Invalid / Missing Value	Major
QR-09	Vital Signs – Visit Date vs Baseline Date	Vital signs visit date does not match any scheduled visit window in the protocol	Protocol Deviation	Major
QR-10	Any Form – Mandatory Field	A field marked mandatory in the CRF is left blank at form submission	Missing Value	Minor
QR-11	Demographics – Ethnicity/Region Text Field	Free-text entry contains inconsistent spelling or abbreviation of the same value (e.g. 'MH' vs 'Maharashtra')	Inconsistent Entry	Minor
QR-12	Baseline – Informed Consent Date	Consent date is recorded after the first study procedure date	Logical Inconsistency / GCP	Critical
QR-13	Vital Signs – Diastolic BP	Diastolic BP is greater than or equal to the Systolic BP recorded at the same visit	Logical Inconsistency	Major
QR-14	Vital Signs – Temperature	Temperature value entered in an ambiguous unit (e.g. 98.6 without °F/°C specified) or outside 90–110°F range	Unit / Range Error	Minor
QR-15	Baseline – Weight Unit	Weight recorded in kg on one visit and lb on another for the same subject without unit conversion flag	Unit Mismatch	Major

QR-16	Any Form – Free-Text Comment Field	Comment field contains a value that contradicts a coded field on the same form (e.g. 'N/A' typed into a numeric field)	Inconsistent Entry	Minor
QR-17	Demographics – Enrolment Number	Enrolment number does not follow the site's approved numbering format (e.g. missing site prefix)	Invalid Format	Minor
QR-18	Baseline – Signature / Date Signed	Form is marked complete but the investigator signature date field is blank	Missing Value / GCP	Critical

Severity legend: Critical = affects subject safety, consent validity, or data integrity and must be queried immediately; Major = affects data quality/analysis and should be queried within standard timelines; Minor = cosmetic or low-impact, batched with routine review.

Query Template Library for Common Data Issues

Standard wording templates below keep queries polite, specific, and actionable. Placeholders in [square brackets] are filled in automatically from the field/record before the query is sent to the site.

Issue Type	Query Template Text
Missing Value	The field [Field Name] on the [Form Name] form for Subject [Subject ID], Visit [Visit Name], is currently blank. Please provide the missing value or confirm if it is not applicable.
Out-of-Range Value	The value entered for [Field Name] ([Value]) on the [Form Name] form for Subject [Subject ID] falls outside the expected clinical range of [Range]. Please verify and correct the value, or confirm if the recorded value is accurate.
Invalid Format	The entry for [Field Name] ([Value]) on the [Form Name] form does not match the expected format ([Expected Format]). Please review and re-enter the value in the correct format.
Duplicate Record	Subject ID [Subject ID] appears more than once in the [Form Name] form. Please confirm which record is correct and identify the duplicate for removal.
Inconsistent Entry	The value entered for [Field Name] ([Value]) is inconsistent with a previous entry for the same subject ([Previous Value]). Please confirm the correct value.
Logical Inconsistency (Date Sequence)	The date recorded for [Field Name] ([Date]) occurs before/after [Reference Date/Field], which is not logically expected. Please verify and correct the date if needed.
Protocol Deviation (Visit Window)	The visit date recorded for [Visit Name] ([Date]) falls outside the protocol-defined visit window of [Window]. Please confirm the reason for the deviation.
Outlier Value	The value recorded for [Field Name] ([Value]) is a statistical outlier compared to the subject's/study's expected distribution. Please confirm if this value is correct.
Unit Mismatch	The unit recorded for [Field Name] ([Value] [Unit]) differs from the unit used in a previous entry for the same subject ([Previous Value] [Previous Unit]). Please confirm the correct value and unit.
Cross-Field Logical Error	The value entered for [Field Name A] ([Value A]) is not consistent with the value entered for [Field Name B] ([Value B]) on the same form. Please review both fields and confirm the correct values.

Missing Signature / Sign-off	The [Form Name] form for Subject [Subject ID] is marked complete, but the investigator signature date is missing. Please complete the sign-off or confirm the correct signature date.
Free-Text vs Coded Field Conflict	The comment entered in [Field Name] ('[Comment Text]') appears to conflict with the coded value selected on the same form. Please clarify which entry is correct.

Style rule: every query is written as a neutral, fact-based statement followed by a clear request for action — never phrased as an accusation of site error.

Sample AI-Generated Query Suggestions

For each detected issue, an AI-generated draft query was produced first, then reviewed and edited by the student before being logged as final. This mirrors real CDM practice, where AI-assisted drafts always pass through human review before being sent to a site.

Issue (from profiling log)	AI-Generated Draft Query	Student-Edited Final Query	Reviewer Comment
Age blank for Subject S-014	Age is missing for this subject. Please fill it.	The Age field is blank for Subject S-014 on the Demographics form. Please provide the subject's age or confirm if this information is unavailable.	Added subject ID and form name for traceability; softened tone.
Systolic BP = 310 mmHg, Subject S-027	BP value of 310 is wrong, please fix it.	The Systolic BP value of 310 mmHg recorded for Subject S-027 at the Baseline Visit is outside the plausible clinical range. Please verify the source document and correct or confirm the value.	Removed judgemental wording ('wrong'); made request neutral and verifiable.
Duplicate Subject ID S-009 in Baseline form	S-009 is duplicated, remove one.	Subject ID S-009 appears twice in the Baseline Visit form. Please confirm which record is the correct entry and identify the duplicate for deletion.	Rephrased as a clarification request rather than a direct instruction to delete.
DOB entered as '15/13/2001' for Subject S-031	DOB is invalid, fix format.	The Date of Birth entered for Subject S-031 (15/13/2001) is not a valid date. Please review and re-enter the correct date of birth in DD-MMM-YYYY format.	Specified the exact invalid value and the required format for clarity.
Heart rate blank for Subject S-042 at Week 4 visit	Heart rate missing, please add.	The Heart Rate field is blank for Subject S-042 at the Week 4 Vital Signs visit. Please provide the recorded value or confirm if the assessment was not performed.	Added visit context and an option to confirm non-performance, avoiding a forced answer.
Diastolic BP (128) higher than Systolic BP (120), Subject S-018	BP values are backwards, please correct.	The Diastolic BP (128 mmHg) recorded for Subject S-018 at the Baseline Visit is higher than the Systolic BP (120 mmHg) at the same visit. Please verify the source document and correct the values as needed.	Replaced informal phrase 'backwards' with a precise description of the inconsistency.

Weight recorded as 70 (unit not specified) for Subject S-055	Please add unit for weight.	The Weight value of 70 for Subject S-055 on the Baseline form does not specify a unit (kg or lb). Please confirm the correct unit and re-enter the value if necessary.	Clarified exactly which value and form the query refers to; kept request short and specific.
Enrolment number 'SUBJ-102' does not match site prefix format 'S03-102', Subject S-102	Enrolment number format wrong.	The Enrolment Number entered for Subject S-102 (SUBJ-102) does not follow the approved site numbering format (e.g. S03-102). Please confirm and correct the enrolment number.	Included both the incorrect value and the expected format so the site can correct it in one pass.

Review pattern observed: AI drafts were consistently short, blunt, and lacked subject/form context; student edits added traceability (subject ID, form, visit), replaced judgemental language with neutral phrasing, and offered the site an option to confirm rather than only an instruction to correct.

Where Query Generation Fits in the CDM Workflow

Query generation is the bridge between error detection and data correction. It does not fix data itself — it formally documents a discrepancy and routes it to the person who can resolve it (usually the site investigator or coordinator). In this training, query generation is the fourth stage in a five-stage cycle applied to the dummy dataset in Castor EDC:

1. Data Collection / Entry	Site enters subject data into Castor EDC forms (Baseline, Demographics, Vital Signs).

2. Data Profiling & Error Detection	Dataset is reviewed for missing values, duplicates, invalid formats, outliers, and inconsistent entries; issues are tagged (as done in the earlier error-classification task).

3. Validation Against Rules	Each tagged issue is checked against the validation rulebook to confirm it genuinely breaks a defined rule (avoids raising false queries).

4. Query Generation (this task)	Confirmed issues are converted into formal, standardised queries using the rulebook and template library; AI-drafted queries are reviewed and edited by the CDM team before release.

5. Query Resolution & Data Correction	Site responds to the query, corrects or confirms the data, and the query is closed and logged for audit trail.

Why this step matters: query generation is the single most repeated task in CDM day-to-day work and is where data quality is actually enforced — a well-written query gets a fast, correct response from the site, while a vague or badly-worded query causes delays, re-queries, and database lock slippage. This is also the stage where AI assistance gives the biggest time saving (drafting standard wording instantly) while human review remains essential for tone, context, and GCP compliance before anything is sent to a site.

Query Lifecycle and Turnaround Targets

Once a query is generated and sent, it moves through a fixed lifecycle with defined statuses. Turnaround targets below are typical illustrative timelines used for training purposes on this dummy dataset, not a specific sponsor's actual SLA.

Status	Meaning	Typical Target
Open	Query has been generated and sent to the site; awaiting site response.	Sent within 1 working day of issue confirmation
Answered	Site has responded with a correction or clarification in the EDC.	Site response within 5 working days
Reviewed	CDM reviewer checks the site's response against the query and the data.	Reviewed within 2 working days of site response
Closed	Response accepted; data corrected or confirmed; query closed with an audit trail entry.	Closed within 1 working day of successful review
Re-Query	Site response is incomplete, unclear, or does not resolve the issue; a follow-up query is raised.	Re-opened same day as review, restarts the cycle

Every status change is logged automatically by the EDC (Castor) with a timestamp and user ID, forming the audit trail that regulators and auditors review to confirm data was cleaned correctly and traceably before database lock.